Problem-Solution Fit

Date	27 June 2026
Team ID	xxxxxx
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Many farmers struggle to identify the most suitable crop for cultivation because they lack knowledge of soil nutrients and environmental conditions. Choosing an unsuitable crop often leads to inefficient use of agricultural resources.

1. CUSTOMER SEGMENT(S) [CS]

- Farmers
- Agricultural Consultants
- Agriculture Students
- Government Agriculture Officers

6. CUSTOMER LIMITATIONS
EG. BUDGET, DEVICES **[CL]**

- Limited knowledge of soil nutrients
- Limited access to agricultural experts
- Budget constraints.

5. AVAILABLE SOLUTIONS
PROS & CONS **[AS]**

Pros: Agricultural experts
Cons: Time-consuming, No instant AI-based prediction

2. PROBLEMS / PAINS
\+ ITS FREQUENCY **[PR]**

- Difficulty selecting the right crop based on soil conditions
- Lack of expert agricultural guidance .

9. PROBLEM ROOT / CAUSE [RC]

- Lack of awareness about suitable crops
- Inadequate soil analysis
- Unpredictable conditions
- Limited access

7. BEHAVIOR
\+ ITS INTENSITY **[BE]**

- Frequently seek advice from expert farmers.
- Search online for suitable crop information

3. TRIGGERS TO ACT [TR]

- Beginning of a new farming season
- Need to improve crop productivity

4. EMOTIONS BEFORE / AFTER **[EM]**

Before: Confused ,Uncertain.
After: Satisfied with crop recommendation

10. YOUR SOLUTION [SL]

OptiCrop – AI-Based Crop Recommendation System uses Machine Learning to analyze soil nutrients and environmental parameters to instantly recommend the most suitable crop through a Flask-based web application. The system helps farmers make informed decisions, improve crop yield, and promote sustainable agriculture..

8. CHANNELS OF BEHAVIOR [CH]

ONLINE: Agriculture Websites ,Mobile Applications ,Search Engines.

OFFLINE: Soil Testing Laboratories ,Farmer Meetings,Agricultural Universities .